Retail Sales & Customer Analysis

Python / Pandas Portfolio Business Report

Dataset: UCI Online Retail | Observation window: 1 December 2010 to 9 December 2011

Prepared from the accompanying Jupyter Notebook. Monetary values are expressed in GBP (£).

1. Project Overview

This project examines historical retail transactions to identify revenue trends, leading merchandise products, international markets, and customer purchasing behavior. The analysis uses Python, Pandas, NumPy, and Matplotlib to prepare the data, calculate business KPIs, and communicate insights to decision-makers.

Business questions: When was revenue strongest? Which merchandise products and countries contributed most? How concentrated is customer revenue? What do average order value, purchasing frequency, and repeat purchasing reveal?

2. Data Preparation

2.1 Data overview and reporting scope

The original UCI Online Retail dataset contains 541,909 transaction-line records. The notebook constructs a cancellation-inclusive transaction dataset of 539,391 rows and a merchandise-oriented subset for product, country, and customer analysis. The observed date range runs from 1 December 2010 to 9 December 2011; the two December periods are incomplete or partial in different ways, and December 2011 ends on the ninth.

2.2 Transaction classification and revenue

Revenue was calculated as Quantity × UnitPrice. Transactions were reviewed for sales, cancellation invoices, zero-price transactions, nonstandard adjustments, and bad-debt entries. Candidate sales and cancellation records were retained for the cancellation-adjusted transaction revenue measure. That measure totals approximately **£9,758,809.99**.

2.3 Missing customer identifiers and duplicate rows

Transactions lacking CustomerID were retained for revenue and merchandise reporting but excluded from customer-level KPIs. Exact duplicate line records were reviewed rather than automatically discarded because identical product lines can be legitimate retail entries; this remains a potential data-quality limitation.

2.4 Special transaction codes and merchandise scope

The notebook standardizes the lowercase StockCode “m” to “M” and separately identifies shipping/postage, financial charges, discounts, samples, manual transactions, and gift vouchers. Those identified StockCodes are excluded from the merchandise-oriented rankings. This is a pragmatic classification: excluding discounts means merchandise revenue is **not fully discount-adjusted**, and some manual entries remain uncertain. The six special categories together account for approximately –£33,898.89 in net transaction revenue.

2.5 Consistent denominators

The report distinguishes cancellation-adjusted transaction revenue from merchandise revenue and from sales-invoice-only customer metrics. Customer KPIs use valid CustomerIDs. Average order value and purchase frequency exclude cancellation invoices from both numerator and denominator.

3. Key Findings and Visualizations

3.1 Monthly revenue trend

November 2011 recorded the highest monthly cancellation-adjusted transaction revenue at **£1,461,756.25**. September and October also showed elevated revenue compared with earlier months. December 2011 recorded £433,668.01, but this reflects only the first nine days and is not a full-month comparison.

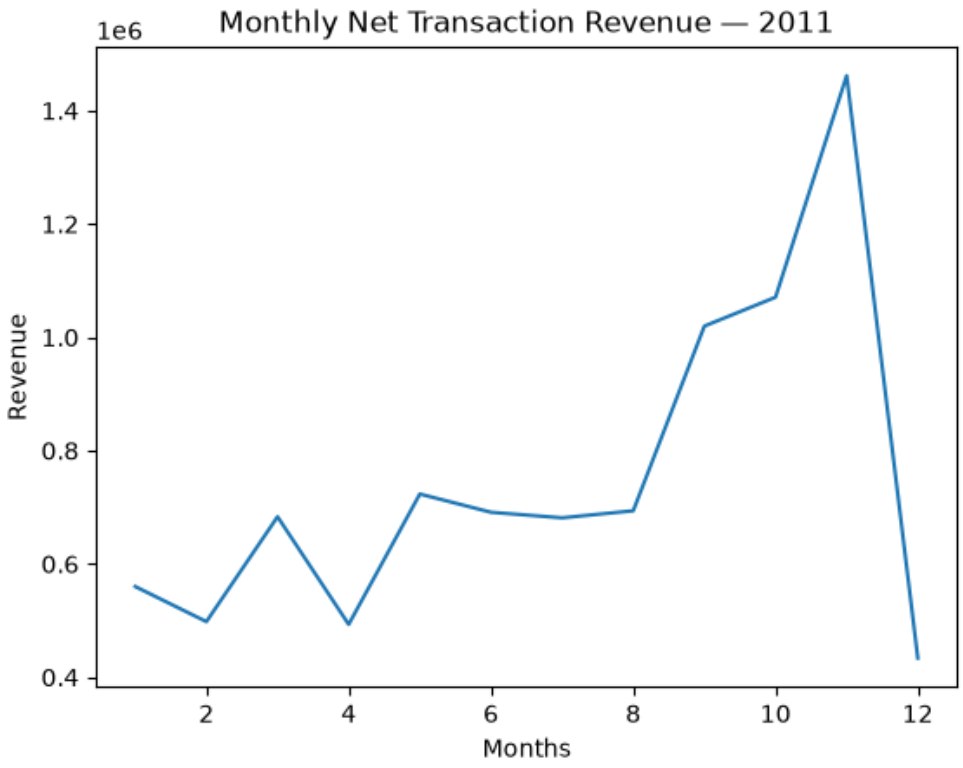


Figure 1. Monthly cancellation-adjusted transaction revenue, 2011 (December is partial).

Business recommendation: Review inventory and operational capacity ahead of the September–November trading period. Avoid treating the December decline as a full-month performance drop.

3.2 Top merchandise products

REGENCY CAKESTAND 3 TIER ranked first by net merchandise-oriented revenue (£164,762.19), followed by PARTY BUNTING (£98,302.98) and WHITE HANGING HEART T-LIGHT HOLDER (£97,715.99). Postage and other identified non-merchandise StockCodes were excluded. Grouping by both StockCode and Description may split products if descriptions vary.

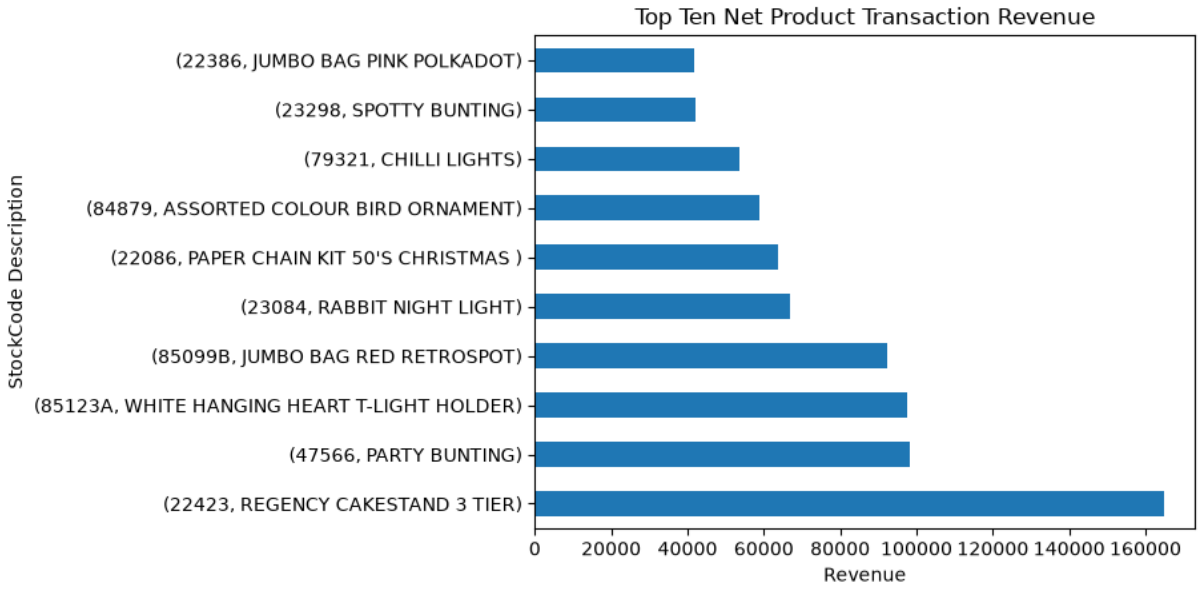


Figure 2. Top merchandise products by cancellation-adjusted revenue.

Business recommendation: Prioritize stock-availability checks for leading merchandise products while validating margins before changing inventory or promotional budgets.

3.3 International market performance

With the United Kingdom excluded to show overseas differences, the strongest international markets were the **Netherlands (£283,479.54)**, **EIRE/Ireland (£259,663.46)**, and **Germany (£200,808.40)**. The United Kingdom was the dominant market in the overall country ranking (approximately £8.30 million).

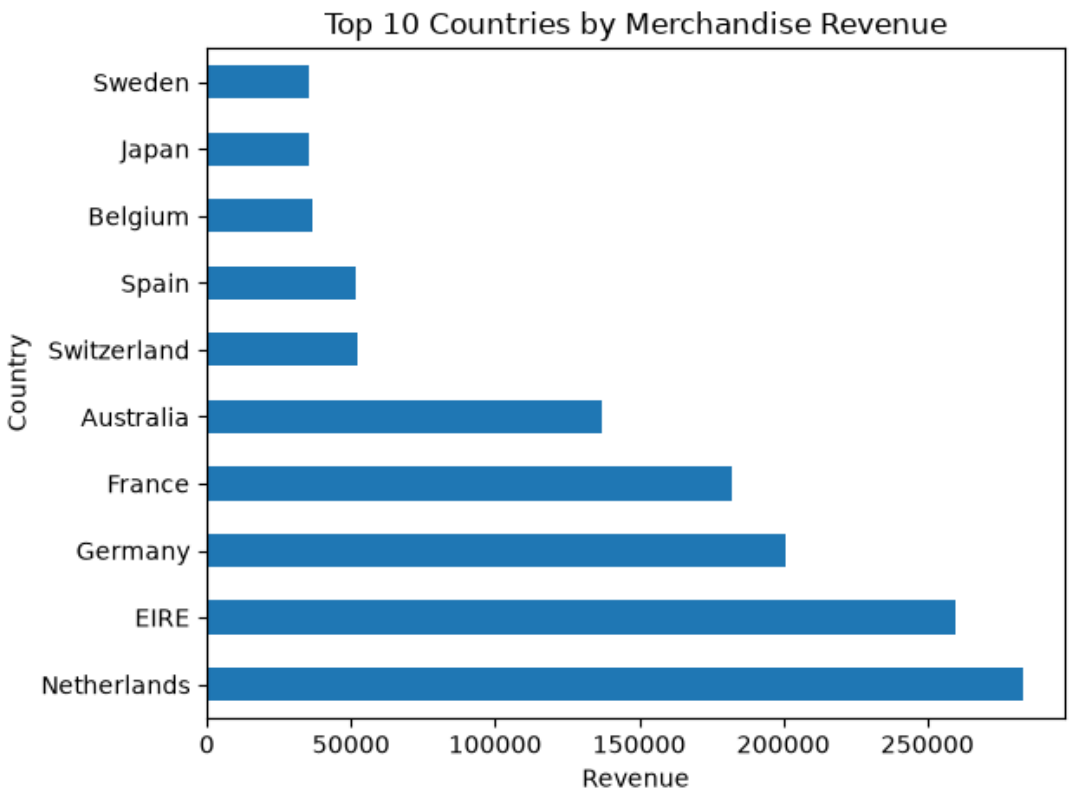


Figure 3. Top international markets by merchandise revenue, excluding the United Kingdom.

Business recommendation: Evaluate the Netherlands, Ireland, and Germany for focused market research, including delivery costs, customer retention, and product demand. Revenue does not establish profitability.

3.4 High-value customers and concentration

Customer 14646 generated the highest identifiable merchandise revenue (£278,778.02), followed by customer 18102 (£259,657.30). The top ten customers generated **£1,370,948.24** out of **£8,286,663.32** in identifiable-customer merchandise revenue, representing **16.54%**.

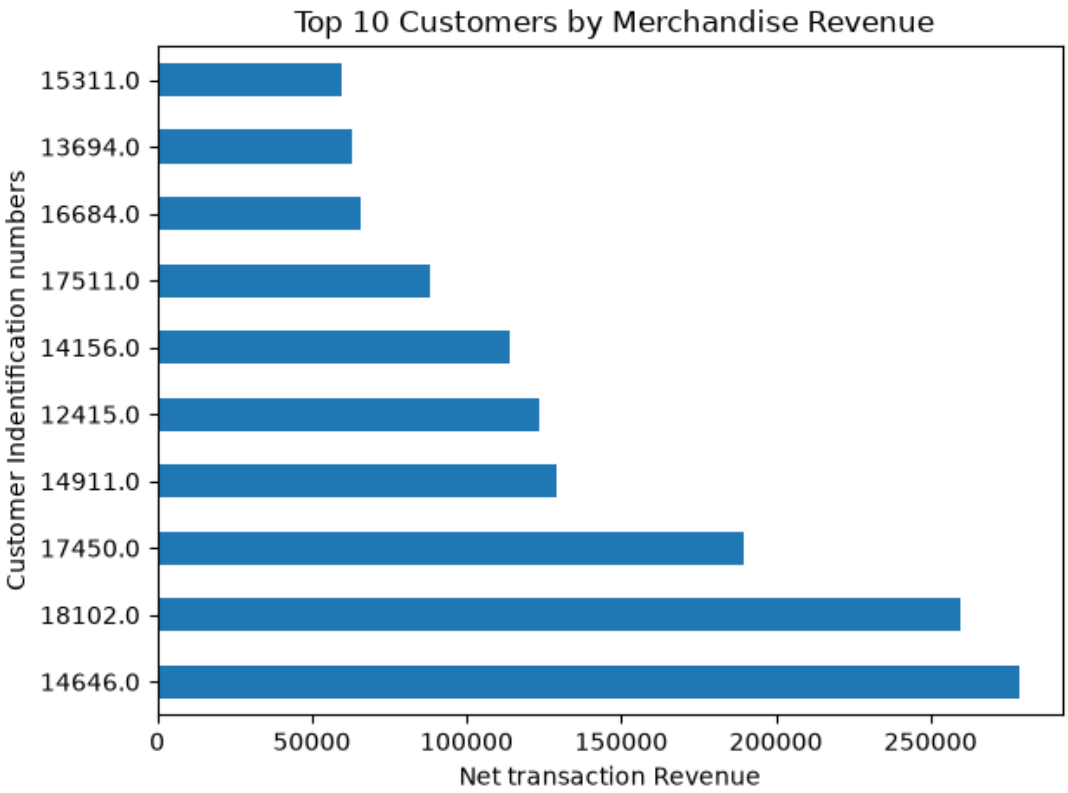


Figure 4. Top ten identifiable customers by merchandise revenue.

Business recommendation: Monitor high-value customer engagement and concentration risk without assuming high revenue equates to high margin or long-term loyalty.

3.5 Customer purchasing behavior

Metric	Result	Definition
Identifiable customers	4,362	Unique valid CustomerIDs in merchandise data
Average revenue per identifiable customer	£1,899.74	Cancellation-adjusted merchandise revenue / identifiable customers
Average sales-invoice value	£476.09	Sales-invoice merchandise revenue / 18,402 unique sales invoices
Average purchase frequency	4.25 orders	Unique sales invoices / 4,334 purchasing customers
Repeat purchasing rate	65.27%	2,829 of 4,334 purchasing customers placed >1 sales invoice

The customer counts differ across metrics because the cancellation-adjusted merchandise population includes customers who may not have a qualifying non-cancellation sales invoice. The repeat rate is an observed repeat-purchase share, **not** a cohort retention measure.

Business recommendation: Explore customer segmentation and repeat-purchase drivers; avoid inferring causality from descriptive KPIs alone.

3.6 Monthly purchasing customers

Unique identifiable purchasing customers rose from **739 in January** to **1,660 in November 2011**, an increase of approximately 124.6%. December showed 614 customers but covers only part of the month. This chart uses sales transactions excluding cancellations.

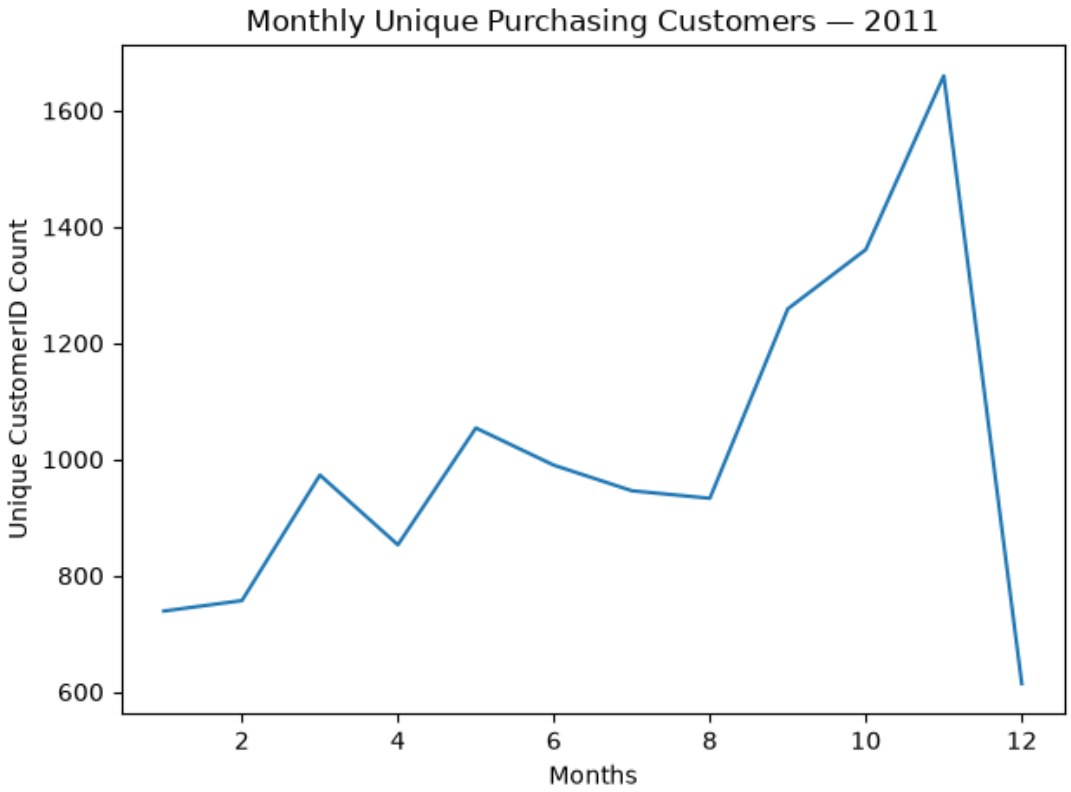


Figure 5. Monthly unique purchasing customers in 2011, excluding cancellation invoices; December is partial.

Business recommendation: Investigate drivers of the late-year increase and plan staffing, fulfillment, and marketing around verified seasonal patterns once additional years are available.

4. Project Conclusion

The analysis identifies a strong late-2011 revenue and customer-activity peak, a small group of leading merchandise products, meaningful overseas revenue opportunities, and substantial observed repeat purchasing among identifiable customers. These findings support targeted follow-up on stock availability, customer engagement, and international market economics.

Key limitations: December 2011 is incomplete; CustomerIDs are missing for some transactions; exact duplicate lines were retained; manual and special transaction codes require business validation; merchandise totals exclude separately classified discounts and charges; and revenue does not measure profit. The notebook is the source for calculations and additional technical detail.

Next analytical steps: Validate transaction classifications with business owners, obtain cost/margin data, compare complete periods across additional years, and perform cohort-based customer retention analysis.